Working with Proportion

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Working with Proportion

What is direct proportion?

- **Direct** proportion
  - As one quantity **increases/decreases** by a certain rate (factor)
  - The other quantity will **increase/decrease** by the same rate
- The **ratio** of the two quantities is **constant**
  - E.g. 2 boxes of cereal is 800 g of cornflakes
  - **Doubling** the number of boxes of cereal (4 boxes) will **double** the amount of cornflakes (1600 g)

How do I solve direct proportion questions?

- **Read** through wordy **direct proportion** questions **carefully**
  - Ensure that you understand the **context** of the question
  - Some questions may tell you the relationship between the two values as a **ratio**
- **Identify** the two quantities involved
  - E.g. Hours worked and pay
- **Find** the **factor** that you will be **increasing/decreasing** by
  - This may be given to you in the question, e.g. 'the amount is tripled'
    - The quantity is multiplied by three
  - Alternatively, find the factor by dividing the 'new' quantity by the 'old' quantity
- **Multiply** the other quantity by this factor to find the required **quantity**
  - E.g. If three times as many hours are worked, the pay will be three times more in total
- Give your **final answer** in **context**
  - **Round** and **give units** where appropriate

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

You may have to round an answer to a whole number, but think carefully about the context of the question!

- Rounding to the nearest whole number is often appropriate
- Sometimes you need to round **up** to the next whole number even if it is not the nearest
  - E.g. If you need 1.3 tins of paint, round the number of tins required up to 2 to ensure that you have enough paint

Help me understand

What is the unitary method?

- The **unitary method** means finding **one of something** (1 **unit** of something)
  - This can be a useful strategy
- For example, find the weight of 7 boxes, if 8 boxes weigh 60 kg
  - Find the weight of **1 box** (1 **unit**) using division
    - 60 kg ÷ 8 boxes = 7.5 kg per box
  - Scale this unit up using multiplication
    - 7.5 kg per box × 7 boxes = 52.5 kg

How do I find which item is the best value?

- A common proportion question is to find which product is the **best value for money**
  - Which is best value for money?
    - 1.5 kg of flour for $0.84
    - or 5 kg of flour for $2.65?
- To do this, compare the prices of 1 kg (1 **unit**):
  - Divide the price in dollars by the weight this buys
    - $0.84 ÷ 1.5 kg = $0.56 per kg
    - $2.65 ÷ 5 kg = $0.53 per kg
  - Make sure you compare prices for the **same size unit** of both items
    - The lowest unit price is the **best** value (0.53 < 0.56)
    - 5 kg of flour for $2.65 is **better value for money** than 1.5 kg of flour for $0.84

Examiner Tips and Tricks

- Start a question by jotting down the **key values** involved
  - You'll be able to pick them out quickly and easily later on

the profit made by the company they work for.
Bonuses are paid at a rate of $250 per $3000 profit the company makes.

(i) Work out the bonus an employee receives if the company makes a profit of $18 000.

(ii) If the company makes less than $600 profit, no bonus is paid.

Find the lowest bonus an employee could receive.

Answer:

(i) Identify the two quantities 'profit' and 'bonus'

Find the factor ('new' ÷ 'old') from the profit

$$\frac{18\,000}{3000} = 6$$

Multiply the bonus by the factor

$$250 \times 6 = 1500$$

Answer in context with units

An employee should receive a bonus of $1500

(ii) We are still working with profit and bonus

The lowest bonus will be when the company makes exactly $600 profit

Find the factor using 'new' ÷ 'old'

$$\frac{600}{3000} = \frac{1}{5}$$

Find the amount of bonus by multiplying by the factor

$$250 \times \frac{1}{5} = 50$$

Answer in context with units

The lowest amount of bonus an employee could receive is $50

What is inverse proportion?

- **Inverse** proportion
  - As one quantity **increases** by a certain rate (factor)
  - The other quantity will **decrease** by the same rate
- This relationship applies vice versa too, if one quantity **decreases** the other **increases**
- E.g. If 2 robots take 15 hours to build a car
  - **Tripling** the number of robots (6) would mean the time taken to build a car would be **divided by 3** (5 hours)

How do I solve inverse proportion questions?

question
    - Some questions may tell you the relationship between the two values as a **ratio**

- **Identify** the two quantities involved
- **Find** the **factor** that you will be **increasing/decreasing** by
    - This may be given to you in the question, e.g. 'the amount is tripled'
    - Alternatively, find this by dividing the 'new' quantity by the 'old' quantity

- **Divide** the other quantity by this factor to find the required **quantity**
- Give your final answer in **context**
    - **Round** and **give units** where appropriate

How do I use the unitary method with inverse proportion?

- This is similar to the unitary method for direct proportion
    - Just remember to do the **opposite** operation when scaling

- For example, 5 workers take 20 hours to complete a job, find how long it would take 8 workers
    - Find the time it would take **1 worker** (1 **unit**)
        - The number of workers decreases
        - Therefore, the time increases
        - 5 × 20 hours = 100 hours

    - Scale this unit up to get the required number
        - The number of workers increases
        - Therefore, the time decreases
        - 100 hours ÷ 8 = 12.5 hours

Examiner Tips and Tricks

- Think about the **context** to determine if a question is **direct or inverse proportion**
    - As the number of robots goes up, the time to build a car comes down (**inverse proportion**)
    - If you buy more boxes of cereal, the amount of cereal also increases (**direct proportion**)

Worked Example

The time taken to fill a swimming pool is inversely proportional to the number of pumps used to pump the water in.
If 3 pumps are used it will take 12 hours to fill the pool.

(i) Work out the amount of time required to fill the pool if 9 pumps are used.

Answer:

Method 1 (Unitary method)

(i)

Find how long it would take one pump

- The number of pumps decreases by a factor of 3
- The time increases by a factor of 3

$$12 \times 3 = 36$$

1 pump takes 36 hours

Find how long it would take 9 pumps

- The number of pumps increases by a factor of 9
- The time decreases by a factor of 9

$$36 \div 9 = 4$$

It takes 4 hours for 9 pumps to fill the swimming pool

(ii)

Find how many pumps are needed to fill the pool in 1 hour

- The time decreases by a factor of 12
- The number of pumps increases by a factor of 12

$$3 \times 12 = 36$$

1 hour requires 36 pumps

Find how many pumps are needed to fill the pool in 6 hours

- The time increases by a factor of 6
- The number of pumps decreases by a factor of 16

$$36 \div 6 = 6$$

6 pumps are needed to fill the pool in 6 hours

Method 2

(i)

Identify the two quantities, 'number of pumps' and 'time (hours)'

Find the factor ('new' ÷ 'old') from the number of pumps

$$\frac{9}{3} = 3$$

Divide the time by the factor

$$12 \div 3 = 4$$

Answer in context with units

It takes 4 hours for 9 pumps to fill the swimming pool

(ii)

We are still working with 'pumps' and 'time'

Find the factor for time using 'new' ÷ 'old'

Find the number of pumps by dividing by the factor

$$3 \div \frac{1}{2} = 6$$

6 pumps are needed to fill the pool in 6 hours

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